BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 053 · Lesson 053 · Graphs: Adjacency Lists and Traversal

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. In a BST, where are keys smaller than the parent?

Q2. What is a graph vertex?

Q3. What does an adjacency list store?

Q4. 024

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

graph — Vertices linked by edges.

adjacency list — Neighbours indexed by vertex.

BFS — Breadth-first search using queue.

visited — Vertices already discovered.

New learning

Discover — An adjacency list records each vertex and its neighbours. For undirected graphs, an edge is represented in both vertices' neighbour lists.

Trace — For BFS, enqueue a starting vertex, mark it visited, and repeatedly dequeue and enqueue undiscovered neighbours.

Evaluate — Traversal order depends on neighbour ordering. A visited set prevents infinite revisiting when the graph contains cycles.

Worked example

Trace BFS from A given neighbours A:[B,C], B:[A,D], C:[A], D:[B].

1. Explain the trace — With B before C, BFS visit order A,B,C,D. Each vertex is marked on enqueue.

We Do · Practise with guidance

Trace a different BFS: X connects Y and Z; Y connects W. Visit from X, Y before Z.

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X:[Y,Z], Y:[X,W], Z:[X], W:[Y]
Queue X → Y,Z → Z,W → W
```

Explain the trace

BFS visit order X,Y,Z,W. FIFO queue processes all distance-one vertices before W.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Write adjacency lists for undirected edges A-B,A-C,B-D. [6]

Question 2

Trace BFS from A in the lesson graph, recording the queue after expanding A and B. [6]

Question 3

Explain why DFS can use a stack while BFS uses a queue and why cycles require visited tracking. [6]

Exit Ticket · Answer before checking

In an undirected graph, A–B is normally stored only under A in a basic adjacency list.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. In the left subtree.

Q2. Node representing a point/entity.

Q3. Each vertex's neighbouring vertices.

Q4. Graph edges A–B, A–C, B–D. BFS from A with alphabetical neighbours: order?

You Do pages · Independent questions

Q1. A:[B,C] [2], B:[A,D] [2], C:[A],D:[B] [2].

Q2. Start [A] [1], after A [B,C] [2], after B [C,D] [2], visited contains A,B,C,D after discovered [1].

Q3. DFS LIFO deep path [2], BFS FIFO frontier [2], visited prevents repeated processing/infinite revisits [2].

Exit Ticket · Model answer

False. Both A and B list one another.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: